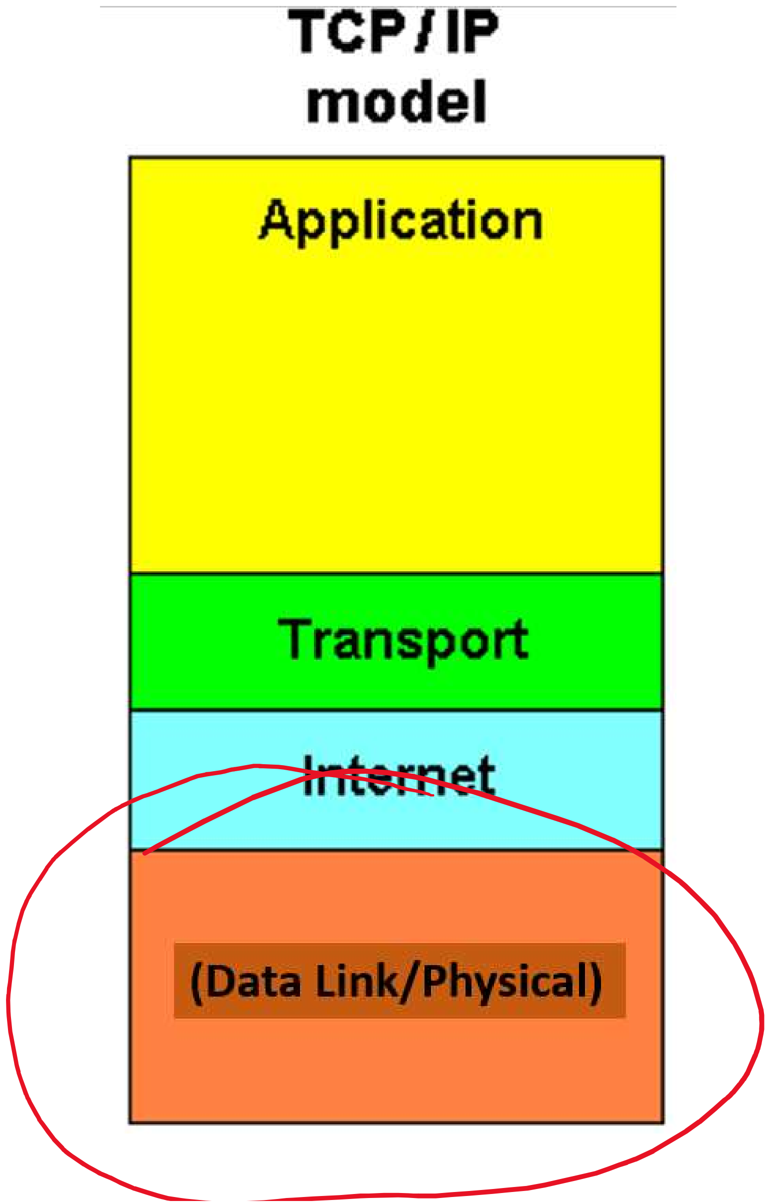
Physical/Data Link Layer



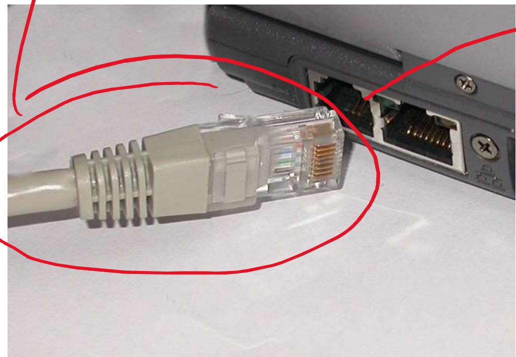
Agenda (1st Part)

- Physical Media
- Purpose of Data Link Layer
- More about topologies
- Data Link Frame
- Ethernet

1. Physical Media

- Networks inherently physical
  - Must have a Physical Connection
- A Network Interface Card (NIC)/Network Adapter connects a device to the network.
- Some devices may contain more than 1 NIC (How many does your Rpi/Arduino have?)
- Can have different performance levels.

RJ 45 Connector



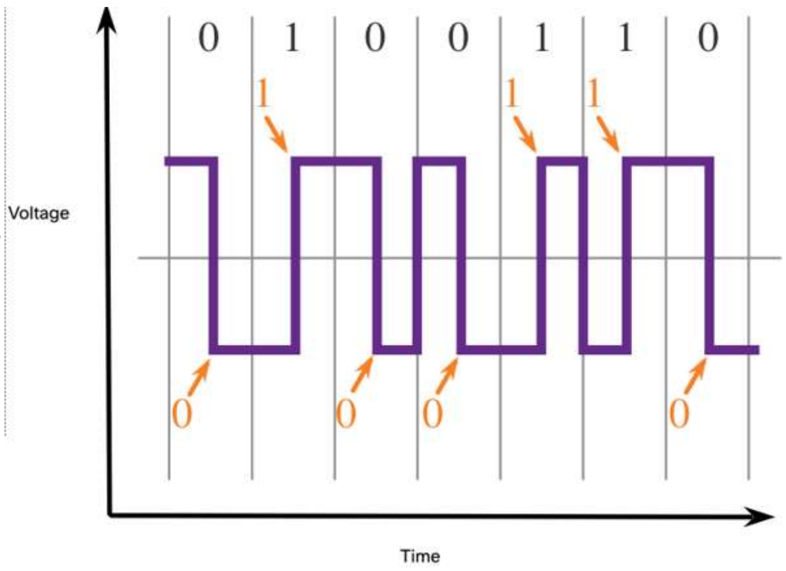
Device NIC

Contains

Wireless NIC

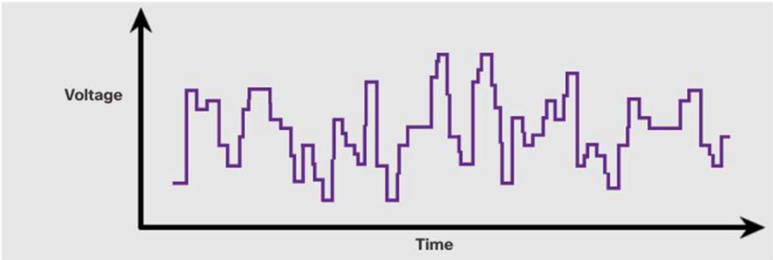
1. Physical Media Characteristics

- NICs/cables are specified in standards associated with the physical layer.
- **Signaling:**
  - the method (voltage, current, RF, light) used to represent binary values on the wire.
- **Encoding:** is how the 1s and 0s will be represented.
  - **Example:** Manchester encoding: A “0” is represented by a high to low voltage transition and a “1” is represented as a low to high voltage transition.

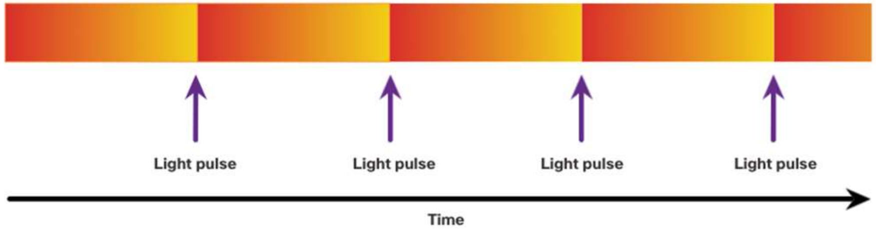


Signaling

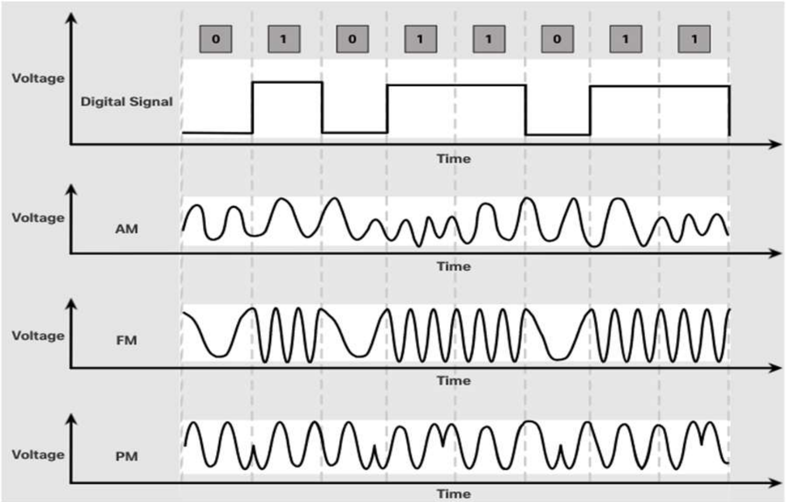
- The signaling method is how the bit values, “1” and “0” are represented on the physical medium.
- The method of signaling will vary based on the type of medium being used.



Electrical Signals Over Copper Cable



Light Pulses Over Fiber-Optic Cable



Microwave Signals Over Wireless

Physical Media: Bandwidth, Throughput, Latency

- Bandwidth is the capacity at which a medium can carry data.
- Digital bandwidth measures the amount of data that can flow from one place to another in a given amount of time; how many bits can be transmitted in a second.
- Physical media properties, current technologies, and the laws of physics play a role in determining available bandwidth.

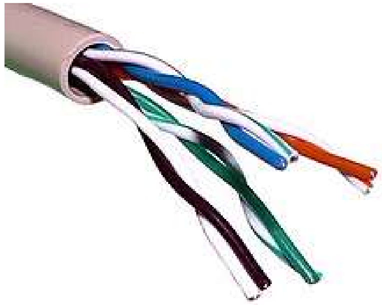
Latency
- Amount of time, including delays, for data to travel from one given point to another

Throughput
- The measure of the transfer of bits across the media over a given period of time

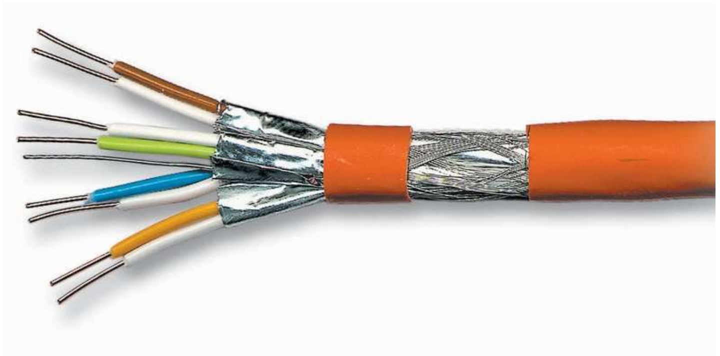
Goodput
- The measure of usable data transferred over a given period of time
- Goodput = Throughput - traffic overhead

Unit of Bandwidth	Abbreviation	Equivalence
Bits per second	bps	1 bps = fundamental unit of bandwidth
Kilobits per second	Kbps	1 Kbps = 1,000 bps = 10^3 bps
Megabits per second	Mbps	1 Mbps = 1,000,000 bps = 10^6 bps
Gigabits per second	Gbps	1 Gbps – 1,000,000,000 bps = 10^9 bps
Terabits per second	Tbps	1 Tbps = 1,000,000,000,000 bps = 10^{12} bps

Physical Media: Copper Cabling



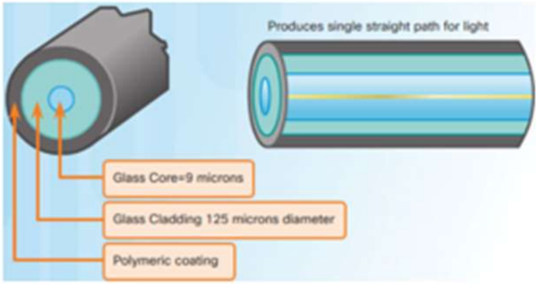
Unshielded Twisted-Pair (UTP)



Shielded Twisted Pair (STP)

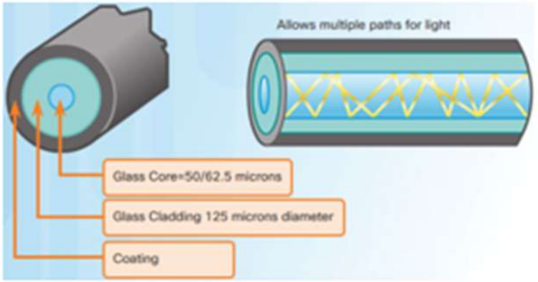
Coaxial Cable

Single-Mode Fiber



- Very small core
- Uses expensive lasers
- Long-distance applications

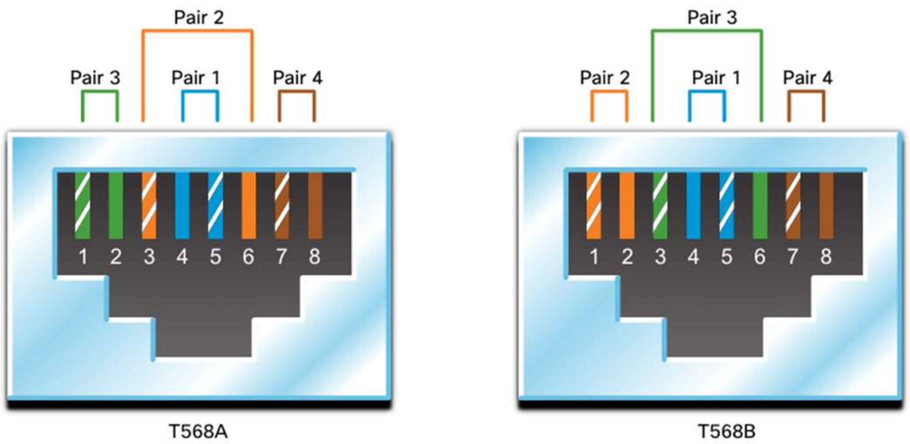
Multimode Fiber



- Larger core
- Uses less expensive LEDs
- LEDs transmit at different angles
- Up to 10 Gbps over 550 meters

UTP Cabling

Straight-through and Crossover UTP Cables



Cable Type	Standard	Application
Ethernet Straight-through	Both ends T568A or T568B	Host to Network Device
Ethernet Crossover *	One end T568A, other end T568B	Host-to-Host, Switch-to-Switch, Router-to-Router
* Considered Legacy due to most NICs using Auto-MDIX to sense cable type and complete connection		
Rollover	Cisco Proprietary	Host serial port to Router or Switch Console Port, using an adapter

Data Link Layer: Main Purpose

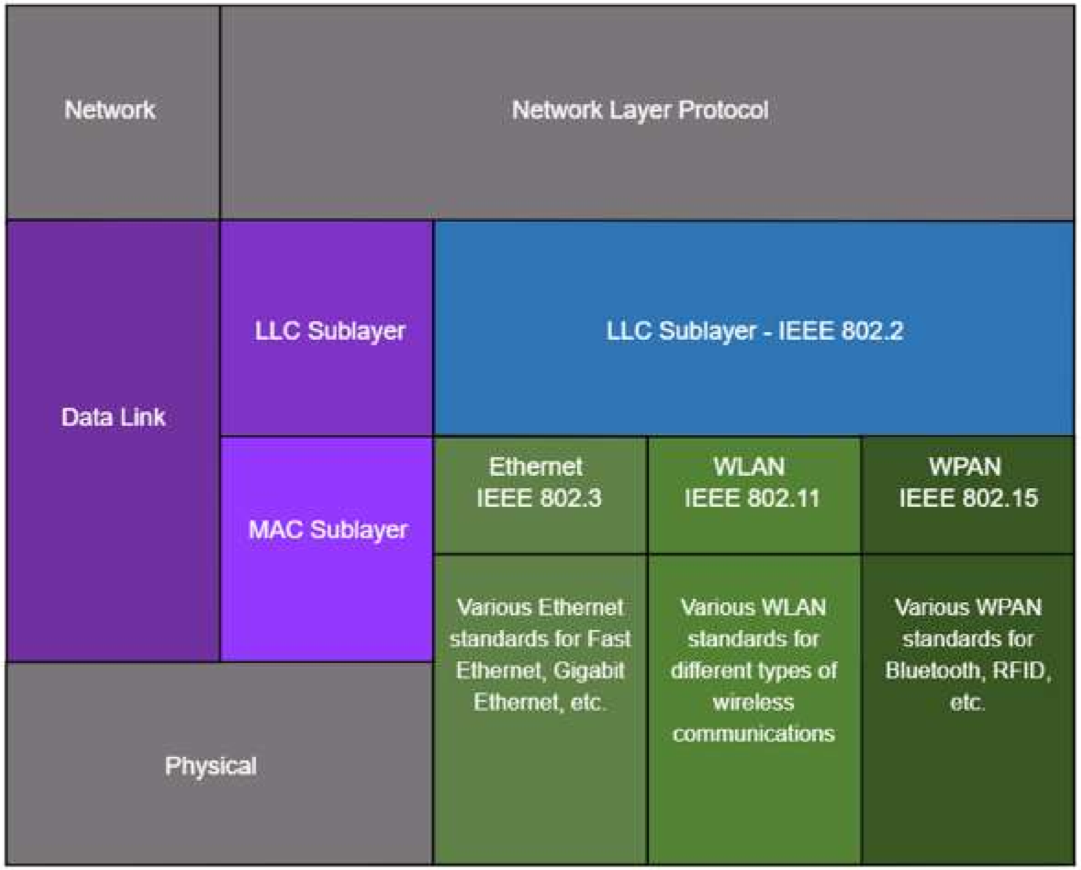
RESPONSIBLE FOR COMMUNICATION BETWEEN END DEVICE NETWORK INTERFACE CARDS(NIC)

PROVIDES ACCESS TO PHYSICAL MEDIA FOR UPPER LAYER PROTOCOLS (I.E. NETWORK LAYER IN TCP/IP)

ERROR DETECTION AND FRAME REJECTION USING

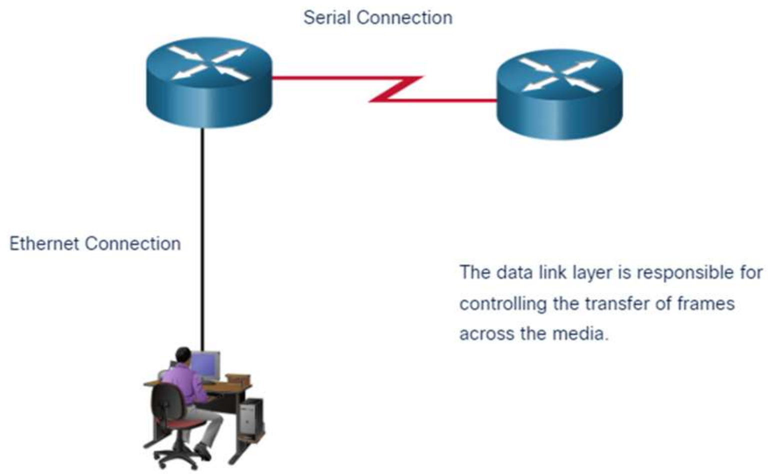
2. Data Link Layer: Standards

- The Data Link Layer has two sublayers. **Logical Link Control (LLC)** and **Media Access Control (MAC).**
  - The LLC sublayer communicates between the networking software at the upper layers and the device hardware at the lower layers.
  - The MAC sublayer is responsible for data encapsulation and media access control.



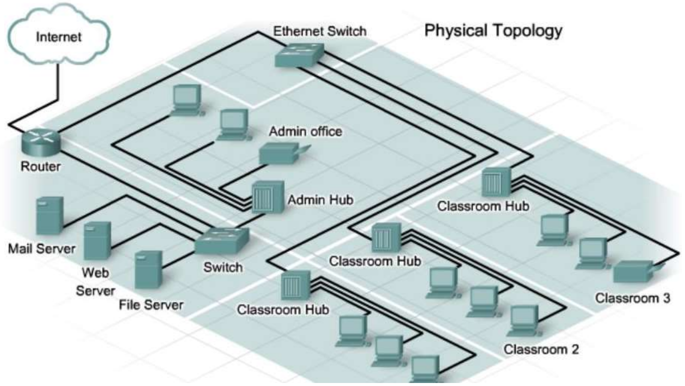
Data Link Layer: Access to Physical Media

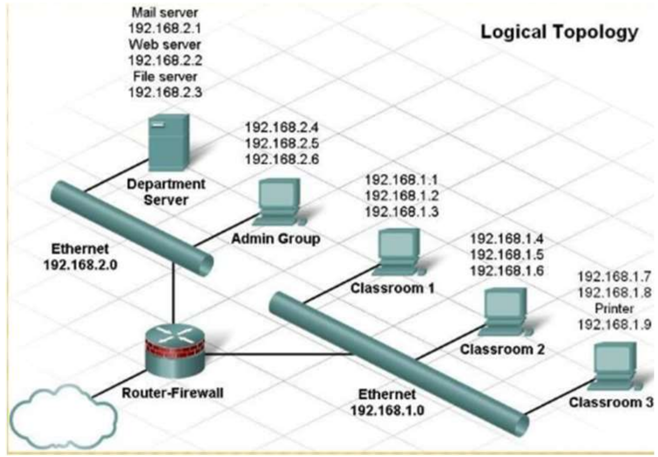
- Communication between nodes might experience many Data Link Layers and Media Types
  - For example, Wifi to Ethernet
- Each "hop", a router will carry out the following functions:
  1. Accepts a frame from the network medium
  2. De-encapsulates the frame to expose the encapsulated packet
  3. Re-encapsulates the packet into a new frame
  4. Forwards the new frame on the medium of the next network segment



2. Data Link Layer: Topologies

- Recap: Network Topology is the arrangement and relationship of nodes and the interconnection between them:
- **Physical Topology:** illustrates physical connections
- **Logical topology:** virtual connections and specifies interfaces and addressing schemes

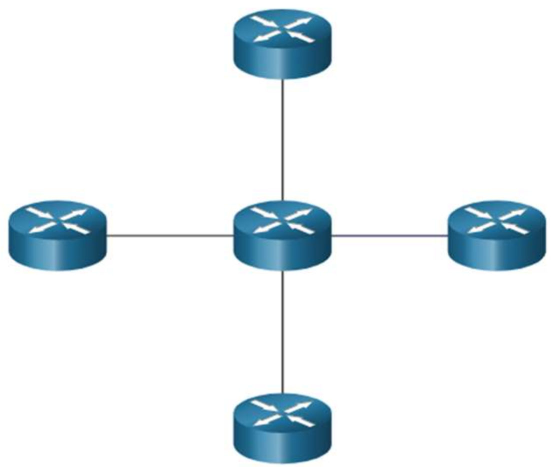


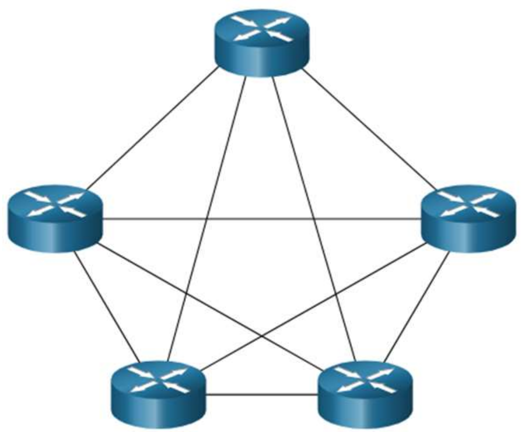


Data Link Layer: WAN Topologies

There are three common physical WAN topologies:

- **Point-to-point** – the simplest and most common WAN topology. Consists of a permanent link between two endpoints.
- **Hub and spoke** – similar to a star topology where a central site interconnects branch sites through point-to-point links.
- **Mesh** – provides high availability but requires every end system to be connected to every other end system.

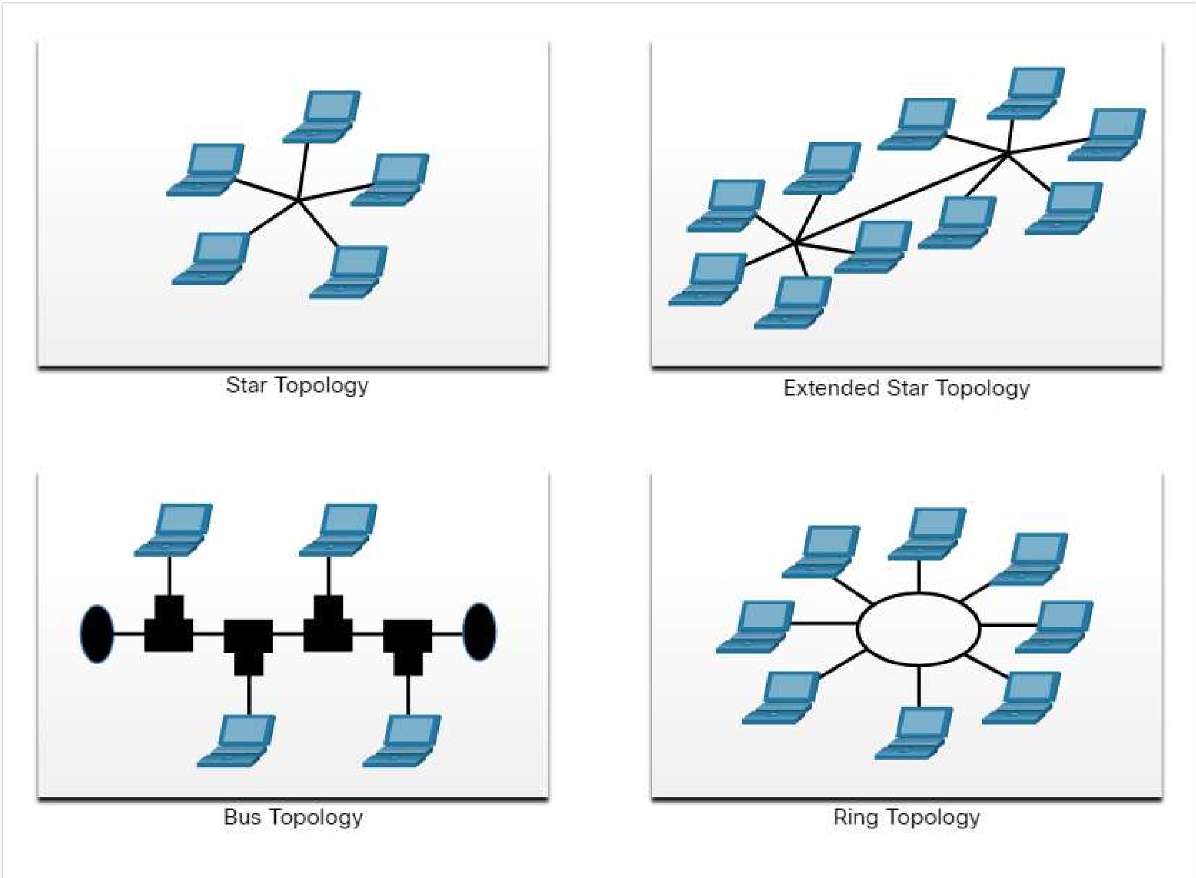




Data Link Layer: LAN Topologies

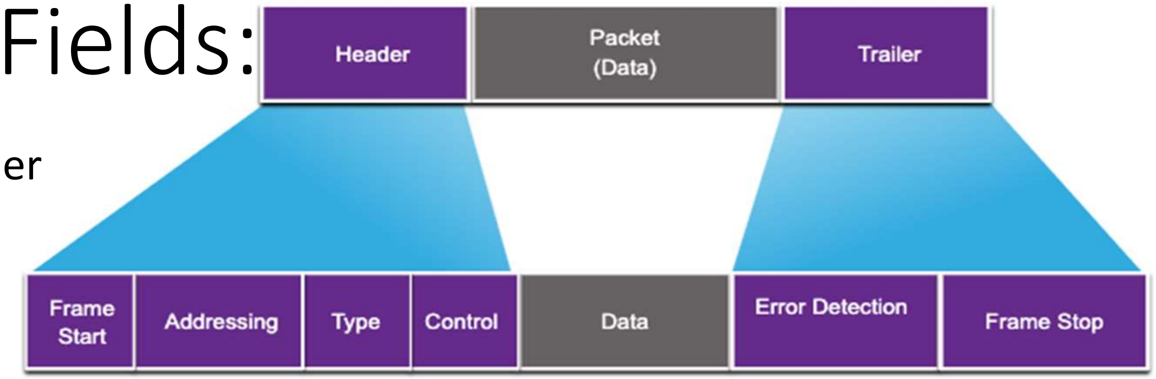
- End devices on LANs typically use star/extended stat topology.
  - Generally easy to scale and troubleshoot.

Physical Topologies



Data Link Frame

Data Link Layer: Frame Fields:



Data is encapsulated by the data link layer with a header and a trailer to form a frame.

A data link frame has three parts:

- Header
- Data
- Trailer

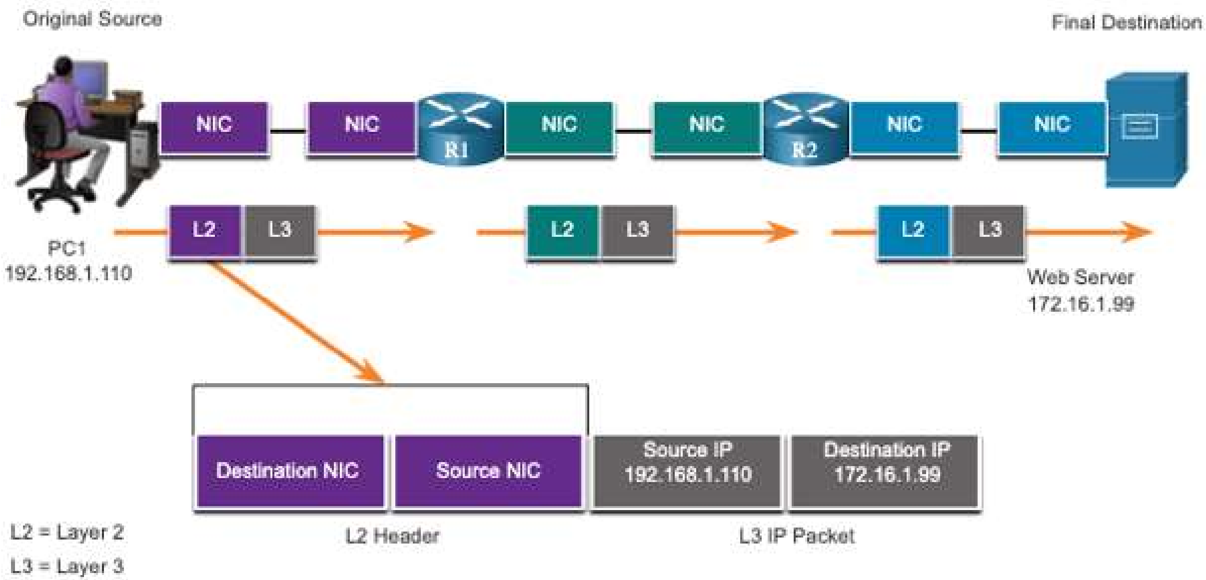
The fields of the header and trailer vary according to data link layer protocol.

The amount of control information carried with in the frame varies according to access control information and logical topology.

Field	Description
Frame Start and Stop	Identifies beginning and end of frame
Addressing	Indicates source and destination nodes
Type	Identifies encapsulated Layer 3 protocol
Control	Identifies flow control services
Data	Contains the frame payload
Error Detection	Used for determine transmission errors

Data Link Layer: Addresses

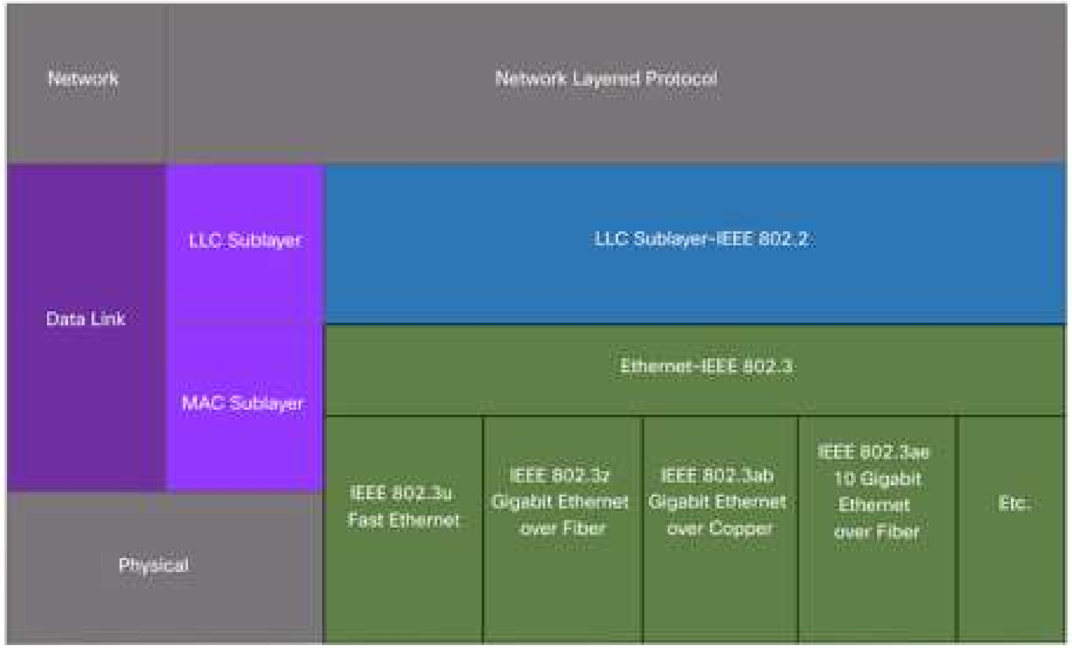
- Also referred to as a physical address or MAC address.
- Contained in the frame header.
- **Used only for local delivery of a frame on the link.**
- **Updated by each device that forwards the frame.**



ETHERNET

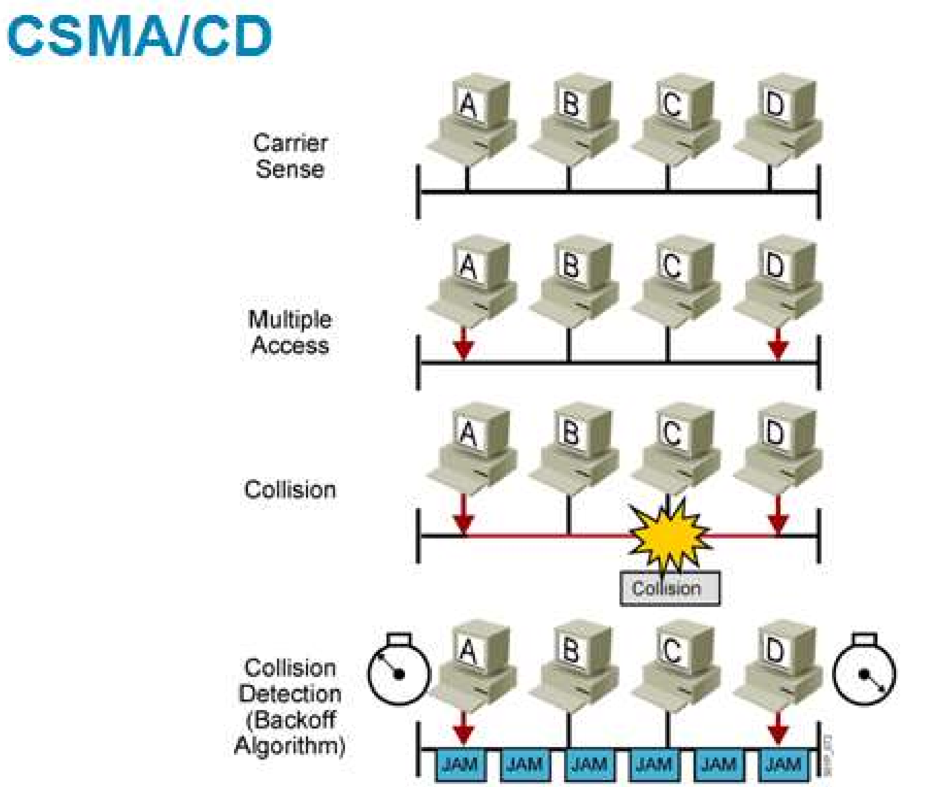
Ethernet: Characteristics

- Operates across Data Link/physical layer
- Provides the following that pure physical layers do not:
  - Connects to upper layers (i.e. network)
  - Provides mechanism to recognise devices
  - Organises bits into frames
- Provides encapsulation into “Frames”
- Ethernet Provides Media Access Control
  - Placement and removal of frames onto media
  - Media access control for ethernet is CSMA/CD
    - All devices on network segment share media
    - All devices receive all frames transmitted on network



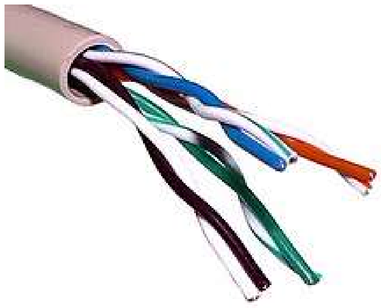
Ethernet: CSMA/CD

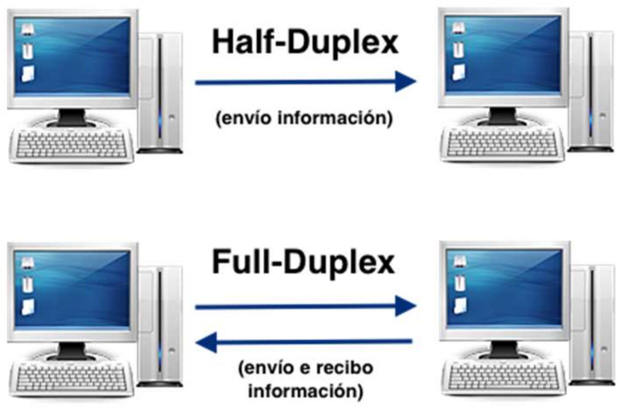
- Some older Ethernet networks use a bus topology or hubs. In shared medium, uses a contention-based access method, carrier sense multiple access/collision detection (CSMA/CD).
- Ethernet LANs of today use switches that operate in full-duplex. Full-duplex communications with Ethernet switches do not require access control through CSMA/CD.



Carrier Sense Multiple Access Collision Detection (CSMA/CD)

Ethernet Collision Detection





- In shared media, only one device can transmit
- More devices on network => more collisions
- Switches reduce collisions
- Switches isolate each port and can send a frame to its destination (if known) rather than every device
- Often uses twisted pair cabling:
  - Allows for one pair for transmission, one for receiving.
- The capability to do both simultaneously is called full duplex
  - No contention for media => no collision domain => No need for CSMA/CD